

364FM.8 MANAGEMENT AND TREATMENT OF CELLULITIS AND ERYSIPELAS FOR ADULTS

Colour code:

- Antibiotics highlighted **red** and **bold** are **penicillin** based. They are contraindicated in patients with a history of penicillin allergy with life-threatening reaction e.g. anaphylaxis, angioedema and/or urticaria.
- Antibiotics highlighted **orange** and **italic** belong to either the cephalosporin or carbapenem groups of antibiotic and should be used with caution in patients with a history of non-severe penicillin allergy e.g. delayed/minor rash. They are contraindicated if serious penicillin allergy e.g. anaphylaxis or angioedema.
- Antibiotics highlighted **green** are considered safe to use in patients allergic to penicillin.

- Definition/Diagnosis** – cellulitis presents as a spreading redness and inflammation of the skin and subcutaneous tissues, often associated with pain/tenderness/swelling, local heat and systemic symptoms (malaise, nausea, pyrexia, rigors) and/or lymphangitis and regional lymphadenopathy.

Erysipelas is usually more superficial with a well-defined raised margin. Cellulitis and erysipelas are most commonly caused by group A or G beta-haemolytic streptococcus (BHS), but *Staphylococcus aureus* is also involved in a significant number of cases, especially when there has been a preceding skin abnormality (boils, folliculitis, impetigo, infected eczema or when there are associated abscesses or ulcers).

Other organisms may be involved in cases of aggressive otitis externa in diabetics (*Pseudomonas sp.*), leg ulcers, pressure sores, following human or animal bites, in diabetics, in IV drug users, children and in the immunosuppressed. Anaerobic infection may be an important factor in necrotising fasciitis and bacterial synergistic gangrene.

- Differential Diagnosis** - includes deep vein thrombosis/thrombophlebitis, varicose/stasis or contact eczema, lipodermatosclerosis, gout, erythema nodosum/panniculitis and localised insect bite.
- Predisposing Factors** - include pre-existing leg ulcers and pressure sores, venous or lymphatic insufficiency, wounds, athlete's foot, fissures on hands and feet, pre-existing dermatitis, obesity, oedema, breast and varicose vein surgery, diabetes and peripheral vascular disease, bites from insects, other animals or humans and foreign travel.
- Complications** - occur in 10% of cases and include tissue necrosis secondary to hypoxia, leg ulcers and chronic oedema/lymphoedema with increased risk of recurrent cellulitis and septicaemia (BHS septicaemia has up to 30% mortality).

5. Management

5.1 Grading of cellulitis – if skin broken, send a swab to microbiology. Draw a line around the margin of the cellulitis and date this to monitor response to antibiotics.

Mild: Localised area of cellulitis without systemic toxicity, complicating factors or other significant co-morbidities can usually be managed in the community with oral antibiotics unless contraindicated.

Moderate: Localised cellulitis with underlying systemic toxicity or co-morbidity, such as peripheral vascular disease, chronic venous insufficiency or morbid obesity. This may complicate or delay resolution of infection. Consider home/outpatient IV antibiotics with provision from ambulatory emergency care unit (AECU), medical day unit (MuDAS) or outpatient parenteral antibiotic therapy (OPAT) (see [BHT Pol 182](#)).

Severe: Significant systemic upset such as acute confusion, tachycardia, tachypnoea, hypotension or unstable co-morbidities that may interfere with a response to therapy or have limb threatening infection due to vascular compromise. Should

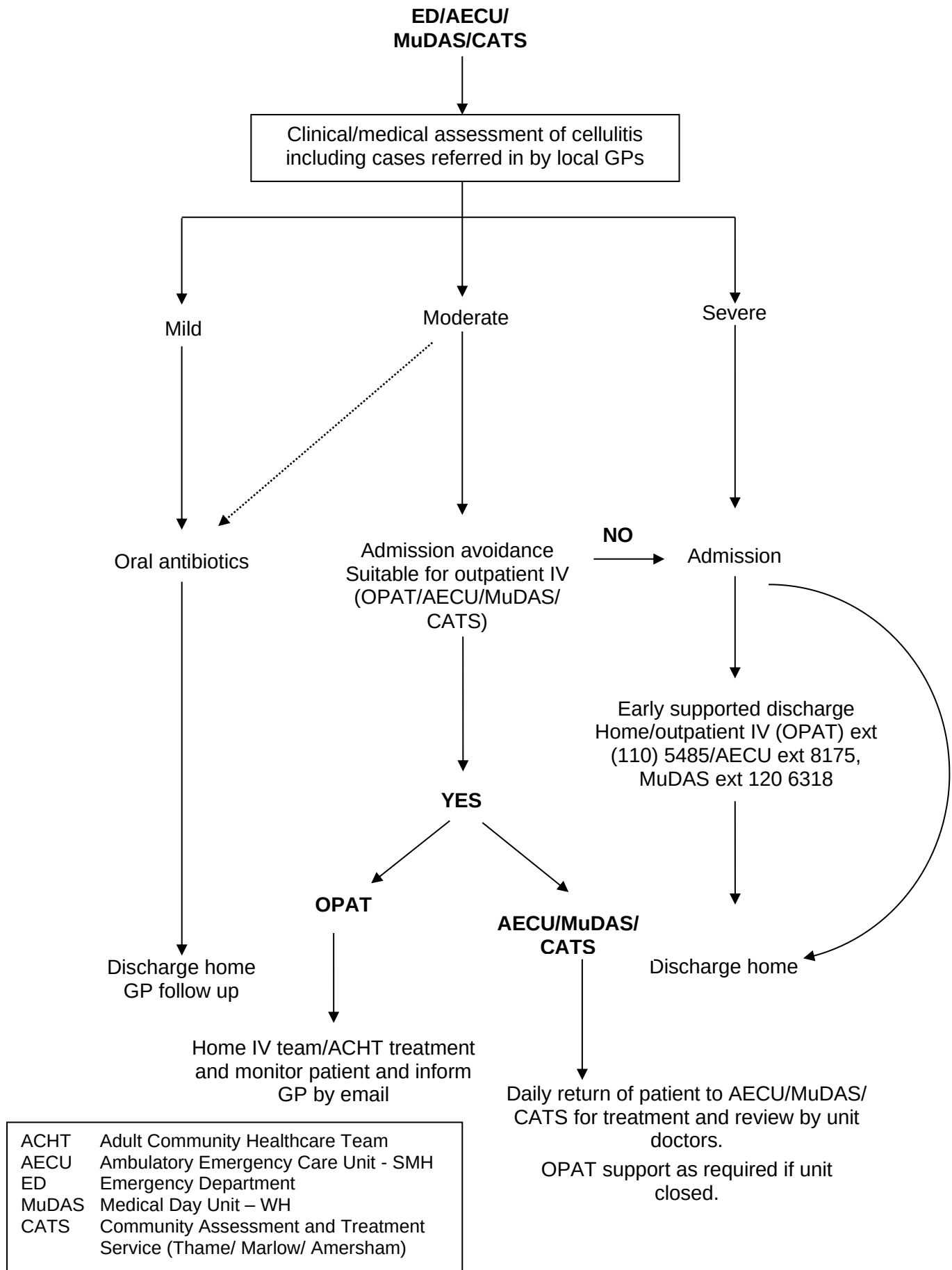
be admitted and commenced on IV antibiotics following blood culture. Other reasons to admit include pregnancy, neonate, resistant or recurrent cellulitis.

Other factors that might need to be considered include extent/severity of cellulitis, site of cellulitis, e.g. cellulitis of face, hands or over joints, tissue necrosis/ulceration, severe lymphangitis, blistering, vomiting or other significant co-morbidities such as immunosuppression, peripheral vascular disease, obesity, alcoholism, diabetes, confusion, frailty, lack of home support, inability to cope.

Patients requiring admission for initial treatment of cellulitis may still become suitable for early supported discharge for home/outpatient IV (as in Moderate section)

Evidence of severe life threatening infection such as necrotising fasciitis (refer to [Urgent Care Sepsis Screening and Action Tool](#) and [Guideline 96FM Antibiotic Therapy for Wound and Soft Tissue Infections including Necrotising Fasciitis](#)). Admit to acute bed and seek specialist advice urgently with early escalation to outreach and referral to Plastics/General Surgical team.

5.2 Algorithm: Cellulitis pathway



5.3 Empirical Antibiotic treatment

Always review choice of therapy with culture and MRSA screen results.

Grade →	Mild	Moderate	Severe If high risk/red flag SEPSIS consider the possibility of necrotising fasciitis and seek specialist advice, e.g. urgent surgical/plastics review/ dermatology/microbiology. Escalate to outreach if necessary.
First line	Flucloxacillin 500 mg - 1 g 6 hourly PO. If patient has had recent course of flucloxacillin in the past 4 weeks then consider second line option below Usual treatment duration: 5 - 7 days A longer course (up to 14 days in total) may be needed based on clinical assessment. However, skin does take time to return to normal, and full resolution at 5 to 7 days is not expected.	Inpatient Flucloxacillin 2 g 6 hourly IV Switch to oral when cellulitis improving or refer to OPAT or ambulatory services. Home/outpatient IV Some patients may be suitable for 8 g/24-hour flucloxacillin infusers. Please discuss cases with OPAT team ext. (110) 5485. <80 years: Ceftriaxone 2 - 4 g 24 hourly IV for 3 - 5 days*, followed by oral switch (see Mild column for oral choice) as per clinical response. >80 years: Teicoplanin 400 mg (if <70 kg) or 600 mg (if >70 kg) 12 hourly for 36 hours then 6 mg/kg (rounded to nearest 200 mg) 24 hourly for 3 - 5 days*, see Guideline 143 followed by oral switch as per clinical response (see Mild column for oral choice). Consider daptomycin[†] 4 mg/kg (increased to 6 mg/kg if associated bacteraemia) 24 hourly IV, for 3 - 5 days* as alternative following a discussion with microbiology. Followed by oral switch as per clinical response (see Mild column for oral choice). Usual treatment duration: 5 - 7 days. May require extension up to 14 days for severe disease or slow response to therapy.	

Grade →	Mild	Moderate	Severe If high risk/red flag SEPSIS consider the possibility of necrotising fasciitis and seek specialist advice, e.g. urgent surgical/plastics review/ dermatology/microbiology. Escalate to outreach if necessary.
Second line/ penicillin allergy	Clarithromycin[#] 500 mg 12 hourly PO or <80 years: Clindamycin 300 – 450 mg 6 hourly PO Usual treatment duration: 5 - 7 days A longer course (up to 14 days in total) may be needed based on clinical assessment. However, skin does take time to return to normal, and full resolution at 5 to 7 days is not expected.	Inpatient <80 years: Clindamycin 450 mg – 600 mg 6 hourly PO (or 600 - 900 mg 6 hourly IV if vomiting or high risk/red flag sepsis) >80 years: Clarithromycin 500 mg IV 12 hourly Home/outpatient IV Teicoplanin 400 mg (if <70 kg) or 600 mg (if >70 kg) 12 hourly IV for 36 hours then 6 mg/ kg (rounded to nearest 200 mg) 24 hourly for 3 - 5 days* see Guideline 143, followed by oral switch as per clinical response (see Mild column for oral choice). Consider daptomycin[†] - see first line treatment for moderate grade in the table above. Usual treatment duration: 5 - 7 days. May require extension up to 14 days for severe disease or slow response to therapy.	Inpatient Vancomycin IV^{\$} (see Guideline 241) or <80 years: Clindamycin 450 mg - 600 mg 6 hourly PO (or 600 mg - 1.2 g 6 hourly IV if vomiting or high risk/red flag sepsis) Once patient stable, can be considered for OPAT or ambulatory services. Total treatment duration: 7 to 14 days.

Grade →	Mild	Moderate	Severe If high risk/red flag SEPSIS consider the possibility of necrotising fasciitis and seek specialist advice, e.g. urgent surgical/plastics review/ dermatology/microbiology. Escalate to outreach if necessary.
MRSA known or suspected Check microbiology results and discuss with micro-biologist	Doxycycline 200mg 24 hourly PO Usual treatment duration: 5 - 7 days. A longer course (up to 14 days in total) may be needed based on clinical assessment. However, skin does take time to return to normal, and full resolution at 5 to 7 days is not expected.	Inpatient Vancomycin IV [#] (see Guideline 241) Home/Outpatient IV Teicoplanin 400 mg (if <70 kg) or 600 mg (if >70 kg) 12 hourly IV for 36 hours then 6 mg/kg (rounded to nearest 200 mg) 24 hourly for 3 - 5 days*, see Guideline 143 followed by oral switch as per clinical response (see Mild column for oral choice). Consider daptomycin [†] - see table on page 4, first line treatment for moderate grade. Usual treatment duration: 5 - 7 days. May require extension up to 14 days for severe disease or slow response to therapy.	Inpatient Vancomycin IV [#] (see Guideline 241) Once patient stable, can be considered for OPAT or ambulatory services. Total treatment duration: 7 to 14 days.

* Draw a line around the margin of the cellulitis and dated this to monitor response to antibiotics; switch to orals as soon as margin recedes. If skin is broken, send swabs to microbiology.

Can be prescribed in patients who have not received oral clarithromycin recently.

† Need to check CK level and drug interactions, e.g. statins.

\$ Do not administer **vancomycin** via a mid-line, switch to **teicoplanin**.

5.4 Non-specific measures - elevation of the affected limb and paracetamol PO 1 g 6 hourly rather than NSAIDs for associated pain/inflammation.

5.5 Advice - from microbiologists, dermatologists (and ophthalmologists, plastics and ENT when face involved).

5.6 Monitor - WCC, CRP, temperature and cellulitic margin. Always check for evidence of athlete's foot between toes, acting as portal of entry, in cases of leg cellulitis. If a fungal infection is found, terbinafine 1% cream applied twice daily is topical treatment of choice.

5.7 Supplementary or alternative antibiotic regimens - may be required:

- Cellulitis secondary to wounds, human and animal bites (see [Guideline 96FM: Antibiotic Therapy for Wound and Soft Tissue Infections including Necrotising fasciitis](#)).
- If exposure to fresh water (ponds, lakes or rivers) at the site of skin break prescribe **ciprofloxacin**¹ 500 mg 12 hourly PO **plus flucloxacillin** 500 mg 6 hourly PO **or** if type 1 penicillin hypersensitivity **ciprofloxacin**¹ 500 mg 12 hourly PO **plus doxycycline** 200 mg 24 hourly PO.
- Cellulitis developing from aggressive otitis externa in diabetics (**ciprofloxacin**¹).
- In diabetics, IV drug users and in the immunosuppressed.

- Microbiological samples taken from the appropriate sites and blood cultures may be helpful in the choice of additional antibiotic.
- **Where anaerobic infection is suspected or in cases of non-responsive cellulitis, metronidazole 500 mg 8 hourly IV/400 mg 8 hourly PO should be added to regimen.**

¹ [Fluoroquinolone Patient Safety Information](#)

6. Facial Cellulitis

(Refer to [Guideline 372FM: Ophthalmic Infections](#) for pre-septal and orbital cellulitis)

a) First line:

Co-amoxiclav 625 mg 8 hourly PO for 7 days

b) Second line/type 1 penicillin hypersensitivity:

<80 years: **Clindamycin** 300 mg 6 hourly PO for 7 days

>80 years: **Clarithromycin** 500 mg PO 12 hourly **plus metronidazole** 400 mg PO 8 hourly for 7 days

7. Recurrent Cellulitis

Do not routinely offer antibiotic prophylaxis to prevent recurrent cellulitis or erysipelas. Give advice about seeking medical help if symptoms of cellulitis or erysipelas develop.

- Predisposing factors should be identified and treated where possible, especially treatment of any underlying oedema/lymphoedema, identifying portals of entry, weight loss, etc.
- Consider referral to dermatology.
- Those at risk could have a 5 day starter supply of either **flucloxacillin** 500 mg 6 hourly, **amoxicillin** 500 mg 8 hourly or **clindamycin** 300 mg 6 hourly PO to take immediately for any new/recurrent attack of cellulitis.
- If there are more than 2 episodes of cellulitis or erysipelas within a year, prophylactic antibiotics may be considered with either **phenoxymethylpenicillin (penicillin V)** 250 12 hourly or **erythromycin** 250 mg 12 hourly PO. Check previous MC&S before initiation. May be continued for several months with interval assessments (minimum 6 monthly) for efficacy and tolerance.
- When choosing antibiotic for prophylaxis take account of any previous microbiological result and previous antibiotic use.

8. Admission Avoidance Protocol for Moderate/Severe Unilateral Cellulitis of Lower Limb

8.1 Referral of cellulitis cases

Patients potentially suitable for admission avoidance therapy can be referred for AECU/OPAT/MuDAS/CATs assessment by any clinician, including GPs, dermatology outpatients, tissue viability nurse, A&E triage. Acceptance or exclusion criteria are used to determine whether an individual may be considered for home/ambulatory IV therapy. If exclusion criteria apply, the patient may still be suitable, but treatment options must be discussed with a microbiologist - see [Appendix 1](#).

References:

1. Fulton R. Doherty L. Gill D, et al. Guidelines on the management of cellulitis in adults. Northern Ireland: CREST. 2005
2. NICE Sept 2019: Cellulitis and erysipelas: antimicrobial prescribing. www.nice.org.uk/guidance/ng141
3. Up to date. Cellulitis and Skin Abscess in Adults. Accessed at https://www.uptodate.com/contents/cellulitis-and-skin-abscess-in-adults-treatment?search=cellulitis§ionRank=1&usage_type=default&anchor=H299498621&source=machineLearning&selectedTitle=2~150&display_rank=1#H299498621
4. IDSA 2014: Practice Guidelines for the Diagnosis and Management of Skin and Soft Tissue Infections.

See also:

[Guideline 96FM Antibiotic Therapy for Wound and Soft Tissue Infections including Necrotising Fasciitis](#)

[Guideline 143 Use of Teicoplanin in Adults](#)

[Guideline 222 Adult and Paediatrics Injectables Guide](#)

[Guideline 241 Intravenous Vancomycin for Adults](#)

[Guideline 249 Assessment of Deep Vein Thrombosis \(DVT\) in the Ambulatory Setting and Anticoagulation Management of DVT and Pulmonary Embolism \(PE\) in Adults \(Age 16 and Over\)](#)

[Guideline 372FM Ophthalmic Infections](#)

[Guideline 733FM Thromboprophylaxis in the Hospital Setting: Reducing the Risk of Hospital Acquired Deep Vein Thrombosis or Pulmonary Embolism](#)

[BHT Pol 182 Adult Outpatient Parenteral Antimicrobial Therapy \(OPAT\)/Home Intravenous \(IV\) Service Policy*](#)

[Sepsis – Urgent Care Sepsis Screening and Action Tool](#)

[Sepsis – Inpatient Sepsis Screening and Action Tool](#)

* BHT users only

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Buckinghamshire Healthcare NHS Trust	

Appendix 1



Buckinghamshire Healthcare
NHS Trust

Cellulitis/Erysipelas Assessment Sheet

Name:

DOB:

Hospital No:

NHS No:

Clinician: Bleep/ext: POD/Location:

Referral received - Date/Time: **Referred from:**

Area affected: Any blisters/wounds:

Swabs taken/date:

Previous cellulitis/erysipelas in the last year: No / Yes - date:

Oral antibiotics history/dates:

Allergies/sensitivities:

Any fevers/rigors: No / Yes Temperature: Initial cellulitis grade:

Previous surgery related to affected limb: No / Yes – describe:

Margins marked: ☐ Any evidence of fungal infection in toes: No / Yes Swabs taken:

U/S scan required to exclude DVT (follow [Guideline 249](#)): No / Yes

Potential Exclusion Criteria for Admission Avoidance under OPAT/AECU/MuDAS/CATS

Tick

Pregnant or breastfeeding

<16 years

Features suggestive of sepsis NEWS >4

Infection involving joints / face

Suppurating wound from bites or foreign objects

Tissue necrosis/ulceration, extensive blistering

Severe lymphangitis

Severe liver or renal failure

Multiple antibiotic allergies

If you answer YES to any of the criteria please ring microbiology consultant on ext 5322 or out of hours 'on-call consultant' via switchboard to discuss IV treatment and patient's suitability for home IV treatment.

If needing home OPAT IV treatment please contact: 07810 181584 for further assessment:
Mon – Sun 09.00 – 15.00.
If out of hours please discuss treatment options with microbiology.

GRADE (mild)

Patient has no signs of systemic toxicity, has no uncontrolled long-term conditions and will be able to take oral antibiotics at home.

(Crest 2005)

GRADE (moderate)

Patient is either systemically ill or systemically well but with a co-morbidity such as peripheral vascular disease, chronic venous insufficiency or morbid obesity which may complicate or delay resolution of their infection.

(Crest 2005)

GRADE (severe)

Patient may have a significant systemic upset such as acute confusion, tachycardia, tachypnoea, hypotension or may have unstable co-morbidities that may interfere with a response to therapy or have a limb threatening infection due to vascular compromise.

(Crest 2005)

If patient has high risk/red flag sepsis or severe life threatening infection such as necrotising fasciitis (refer to [Sepsis Urgent Care Screening and Action Tool](#)).

(Crest 2005)

Home oral antibiotic therapy unless contraindicated.

[Guideline 364](#) for treatment options.

Refer back to GP for follow-up.

Consider non-acute IV therapy.

Refer to [OPAT Policy](#) (BHT Pol 182)/AECU/MuDAS/CATS
Ensure exclusion criteria considered and if necessary, microbiology input achieved.
[Guideline 364](#) for treatment options.

Admit to acute bed and consider AECU/MuDAS/OPAT when stable.

[Guideline 364](#) for treatment options +/- [Sepsis Urgent Care Screening and Action Tool](#).

Refer to microbiology if further input required.

Admit to acute bed and seek specialist advice, e.g. plastics/dermatology/microbiology urgently.

[Guideline 364](#) for treatment options +/- [Sepsis Urgent Care Screening and Action Tool](#).

Refer to microbiology if further input required.

Confirmed Cellulitis Grade: Signed: Date: